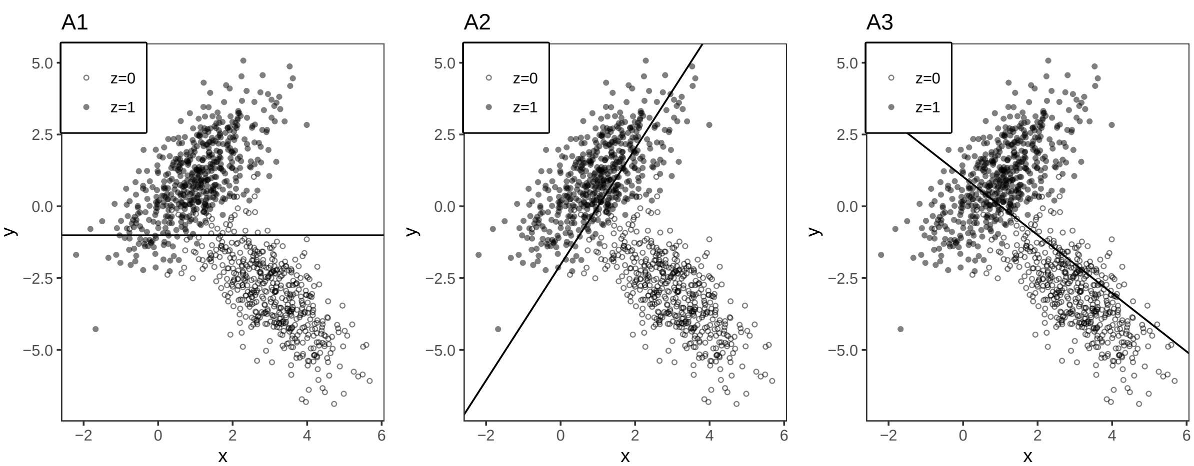
Midterm 2 Practice Problems

Problem 1

In the following three parts, choose the figure with the appropriate regression line(s).

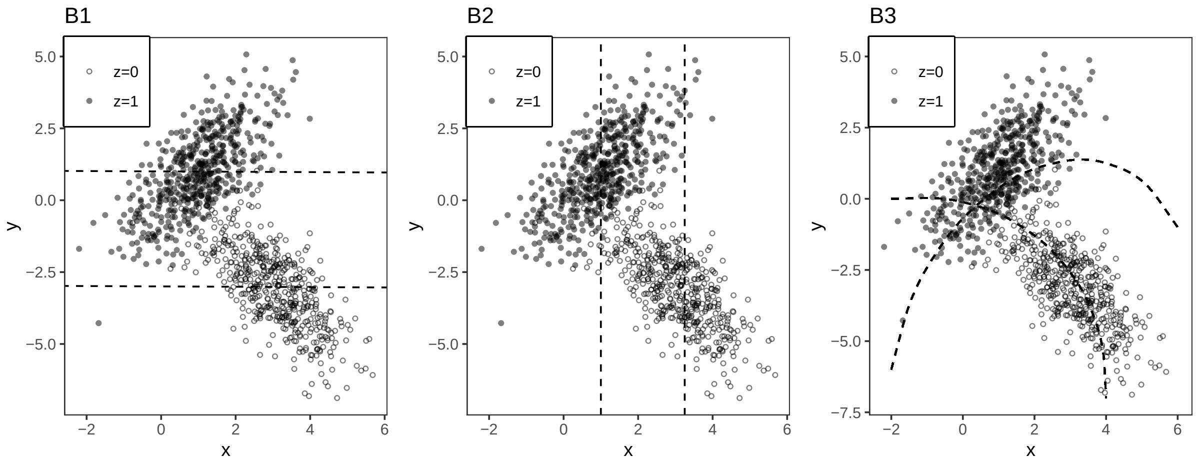
Part A

Choose the figure with the solid line that represents the simple regression line ($y \sim x$).



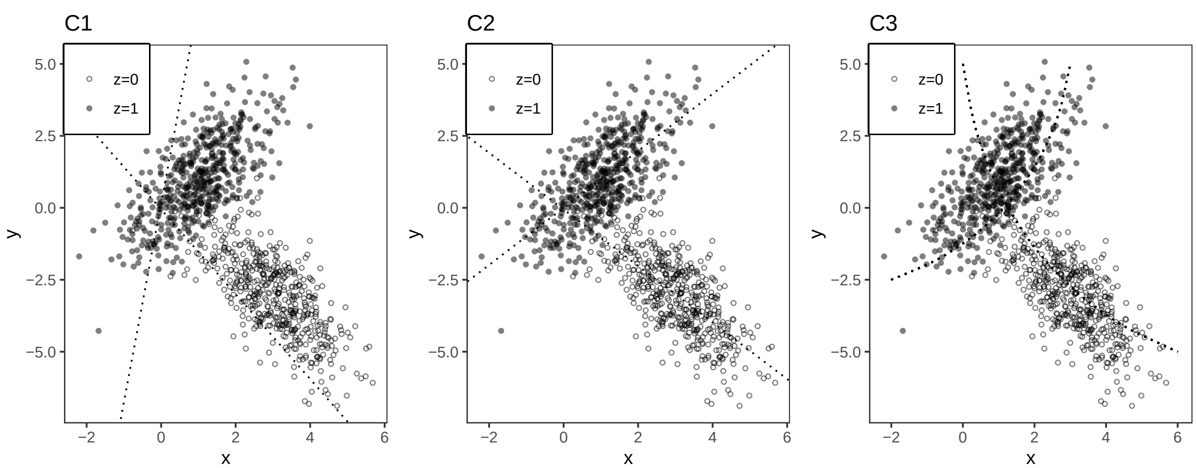
Part B

Choose the figure with the two dashed lines associated with the additive multiple regression ($y \sim x + z$).



Part C

Choose the figure with the two dotted lines associated with the interactive multiple regression ($y \sim x * z$).



Problem 2

Suppose we have a sample $Y_1 \ldots Y_n$ stored on an **R** vector Y of length n. We run this code.

```
B=10000
Y.bar = rep(NA,B)
for(b in 1:B) {
  Y.bar[b]= mean(sample(Y,size=n,replace=TRUE))
```

```
}
```

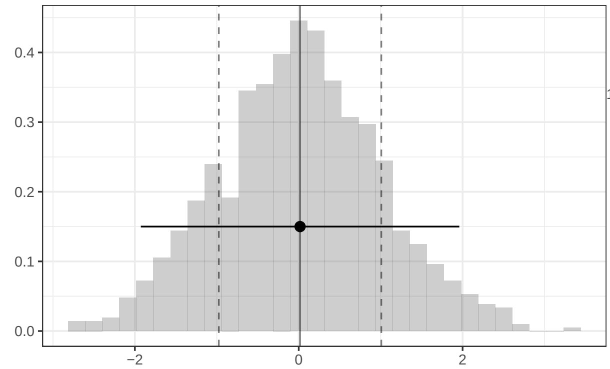


Figure 1: Histogram of the vector Y's elements

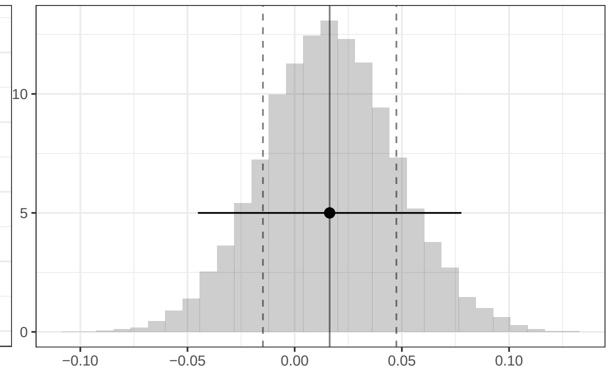


Figure 2: Histogram of the Y.bar's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y) - 1.96* sd(Y)/sqrt(n)
mean(Y) + 1.96* sd(Y)/sqrt(n)
```

Part b

How could you calculate roughly the same thing as in part a using 'sd(Y.bar)' and not 'sd(Y)'?

Problem 3

Consider the following `R` code:

```
1. lm(residuals(lm(y~z)) ~ residuals(lm(z~x)))
2. lm(y ~ residuals(lm(z~x)))
3. lm(residuals(lm(y~z)) ~ residuals(lm(x~z)))
4. lm(residuals(lm(y~z)) ~ x)
5. lm(residuals(lm(y~x)) ~ residuals(lm(x~z)))
6. lm(residuals(lm(y~x)) ~ z)
7. lm(residuals(lm(y~x)) ~ residuals(lm(z~x)))
8. lm(y ~ residuals(lm(x~z)))
```

a) Choose one of the above that will produce the coefficient on x from the multiple regression of $y \sim x + z$

b) Choose one of the above that will produce the same residuals as the residuals from the multiple regression of $y \sim x + z$

Problem 4

We take a random sample of n individuals and record their years of education and income. Y_i is an individual i's income and X_i is that individual's number of years of education. Define

$$\hat{\mu}(16) = \frac{1}{\sum_{i=1}^{n} 1_{16}(X_i)} \sum_{i=1}^{n} 1_{16}(X_i) Y_i$$

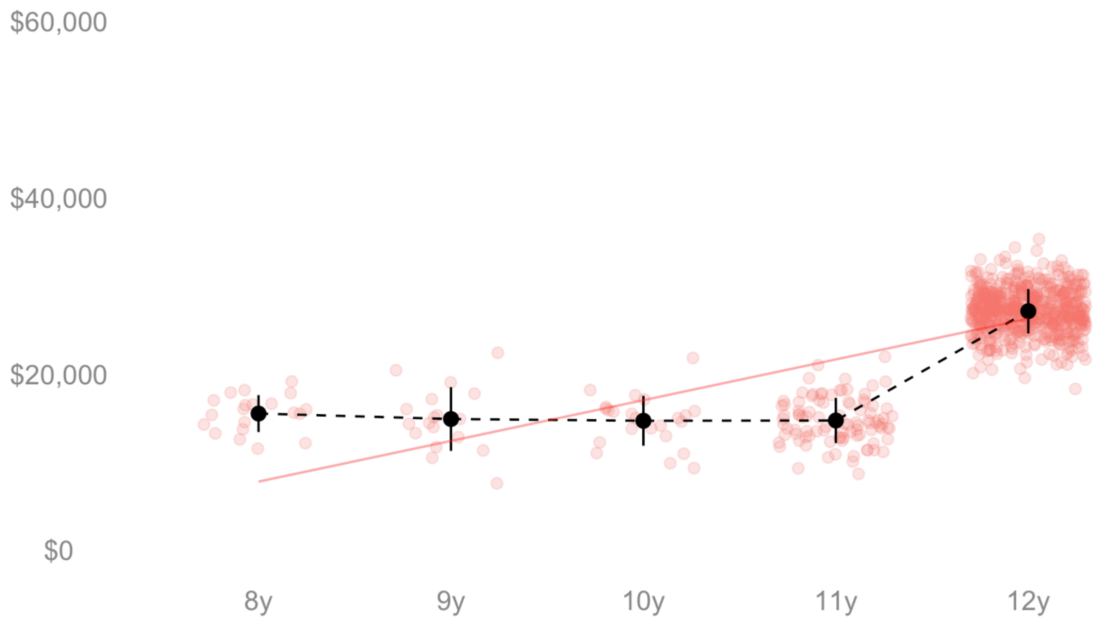
as the subsample mean income among those with 16 years of education. Define the indicator function for $x = 16$ as

$$1_{16}(X_i) = \begin{cases} 1 & if\ x = 16 \\ 0 & otherwise \end{cases}$$

Show that $\hat{\mu}(16)$ is an unbiased estimator of $\mu(16)$, the subpopulation mean income for those with 16 years of education. Justify your steps.

Problem 5

In the below visualization, the x-axis is an individual's years of education, the y-axis is income, the red line is the least squares line, and the black dots indicate the mean income within each year of education. You have three options to summarize the change in income across an additional year of education: the average over years, the average weighted by the proportion of individuals with the years of education P_x, and the slope of the least squares line. Rank these options smallest to largest regarding the magnitude of the estimate you expect and explain your reasoning.



Income and Years of Education

Problem 6

Suppose we want γ_1 from $y_i = \gamma_0 + \gamma_1 x_i + \gamma_2 w_i + e_i$ but we run the regression $lm(y \sim x)$ without w. We have de-meaned our variables x_i and w_i such that $\bar{x} = 0$ and $\bar{w} = 0$. Will the coefficient on x from $lm(y \sim x)$ be equal to γ_1? Derive what your coefficient on x will be in terms of γs and justify your steps.